

Algebra General Exams 1977-2002

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Algebra General Exam

Algebra General Exam

1. Show (by Sylow theory) that a group of order 72 cannot be simple. List all abelian groups of order 72.
2. Let G be an infinite group containing an element $x \neq 1$ having only finitely many conjugates. Prove that G is not simple.
3. Let $f(x) = x^4 - x^2 - 2 \in \mathbb{Q}[x]$. Find the splitting field F of $f(x)$ over $\mathbb{Q}$, determine its Galois group, and find all subfields of F .
4. Let $k \subseteq K \subseteq L$ and let $\alpha \in L$. Assume that K/k is finite algebraic.
 - (a) Show that K/k separable $\iff K(\alpha)/K(\alpha)$ separable.
 - (b) Show that K/k normal $\implies K(\alpha)/K(\alpha)$ normal.
 - (c) Show that if K/k is Galois, then so is $K(\alpha)/K(\alpha)$ and compare the Galois groups $G(K/k)$ and $G(K(\alpha)/K(\alpha))$.
5. What are necessary and sufficient conditions on the commutative integral domain R in order that the polynomial ring in one indeterminate, $R[x]$, be
 - (a) a P.I.D.?
 - (b) a UFD?
 - (c) a Noetherian ring?
6. Let R be a commutative ring with unit and I an ideal of R . Consider the R -module R/I .
 - (a) Show that if I is a prime ideal, then R/I is indecomposable.
 - (b) Show that R/I is a simple module if and only if I is maximal.
7. Let T be a linear transformation of $\mathbb{R}^5$ having characteristic polynomial $T^5 - T$. Find its possible rational canonical forms over $\mathbb{R}$ and its possible Jordan canonical forms over $\mathbb{C}$.

ALGEBRA GENERAL EXAMINATION

ALGEBRA GENERAL EXAMINATION

Algebra General Exam, January 28, 1995

ALGEBRA GENERAL EXAMINATION

January 28, 1995

This examination is closed book. Please write your answers out neatly and include all your scratch work. The time allowed is three hours. and there are eight problems.

1. In keeping with tradition, let G be a group of order 1995. Show that G must be solvable. (Although you needn't prove them here, you may use only theorems you are able to prove!)
2. Explain the connection between the Fundamental Theorem of Arithmetic and the Jordan-Hölder Theorem applied to cyclic groups.
3. Let K and L be finite extensions of a field F , both contained in a field E . Let KL be the set of sums of products of members of K and L .

Explain why KL is a subring of E that is finite dimensional over F . From that, show that KL is a field and, in fact, the smallest subfield of E containing K and L .

4. List all the possible Jordan canonical forms for a matrix over the complex field whose characteristic polynomial is $x^8 - x^4$ and whose minimal polynomial is $x^6 - x^2$.
5. Give definitions of the terms "maximal subgroup" and "minimal subgroup" - it is not assumed that you have seen these terms explicitly. Then from your definitions, prove the following facts:
 - a) A minimal subgroup must be cyclic of prime order.
 - b) If a subgroup has prime index, it is a maximal subgroup.
 - c) If a subgroup is both maximal and normal, it has prime index.
 - d) A subgroup of an Abelian group is maximal if and only if it has prime index.

Now find the maximal and minimal subgroups of $\mathbb{Z}$.

6. Prove that any proper homomorphic image of a principal ideal domain that is an integral domain must actually be a field.

Use this result to show that $F[x, y]$, F a field, is not a principal ideal domain.

7. Show that $x^4 - 2x^2 - 2$ is irreducible over $\mathbb{Q}$. Label its roots in some way and display the Galois group as a set of permutations of the roots.
8. Give an example of a polynomial $f(x)$ in $\mathbb{Q}[x]$ having all these properties:
 - a) the degree of $f(x)$ is 4 ;

- b) $f(x)$ has no rational roots;
- c) $f(x)$ has no repeated factors in $\mathbb{Q}[x]$;
- d) the Galois group of $f(x)$ over $\mathbb{Q}$ is cyclic of order 2 .

Algebra Comprehensive Exam

Algebra Comprehensive Exam

August, 1995

1. Describe a nonabelian group of order 55 as semidirect product, and, using your description, explicitly give generators and defining relations for the group.
2. Let $\rho_1 : G \rightarrow \text{Aut}(V)$, $\rho_2 : G \rightarrow \text{Aut}(W)$ be linear representations of a finite group G . Let $L : V \rightarrow W$ be a linear map. Show that the linear map $L' : V \rightarrow W$, defined by

$$L'v = \left(\frac{1}{|G|} \right) \sum_{g \in G} \rho_2(g^{-1}) L \rho_1(g) v, \quad v \text{ in } V,$$

is an intertwiner of the representations ρ_1 and ρ_2 , i.e. $L' \rho_1(g) = \rho_2(g) L'$, for all $g \in G$.

3. Let G be the Galois group of $x^4 - 4$ over the field $\mathbb{Q}$ of rational numbers and let F be the splitting field of $x^4 - 4$. Find G and describe the action of each of its elements on F . Also find all subfields of F .
4. If T is a diagonalizable linear operator on a finite dimensional vector space V , and if W is a T -invariant subspace of V , prove there is a T -invariant subspace U such that $V = W \oplus U$.
5. Factor the polynomial $x^9 - x$ in $F_3[x]$ into irreducible factors. (F_3 is the field with three elements.)
6. Assume T is a linear operator on a 10-dimensional real vector space V with minimum polynomial $(x^4 - 1)^2$, and let k denote the maximum number of linearly independent eigenvectors of T in V . What are the possible values of k ? Justify your answer.
7. Find the order of the group $\text{GL}(n, F_p)$, and describe a p -Sylow subgroup. (p is a prime number, and F_p is the field with p elements.)
8. A representation of the symmetric group S_3 in the vector space $\mathbf{R}^3$ is defined by letting a permutation in S_3 act on a vector in $\mathbf{R}^3$ by permuting the coordinates of the vector. Express $\mathbf{R}^3$ as a direct sum of irreducible S_3 -invariant subspaces. Prove that your direct summands are irreducible S_3 -invariant subspaces.
9. List the conjugacy classes of finite subgroups of the orthogonal group $O(2, \mathbf{R})$, and prove your list is correct.
10. The space $M_n(\mathbf{R})$ of $n \times n$ real matrices is a real inner product space with inner product given by: $\langle A, B \rangle = \text{trace}(AB^t)$, where B^t denotes the transpose of the matrix B . (You don't need to prove this.) Let P be an invertible matrix and T the linear operator on $M_n(\mathbf{R})$ defined by $T(A) = P^t A P$. Denote the adjoint of T by T^* . Prove that $T^*(A) = P A P^t$, and find necessary and sufficient conditions on the matrix P that $T = T^*$. Justify your answer.

Algebra General Exam, August 26, 1995

ALGEBRA GENERAL EXAMINATION

August 26, 1995

This algebra examination is closed book. Please write your answers out neatly and include all your scratch work. The time allowed is three hours, and there are seven problems altogether.

1. Let G be a finite group of permutations of the finite set X . For $x \in X$, let

$$G_x = \text{Stab}(x) = \{g \in G : gx = x\}.$$

Suppose $|X| = [G : G_x]$ for some $x \in X$. Show this holds for all $x \in X$.

2. Let G be a group of order 1001. Show:

a) G is not simple;

b) G is Abelian;

c) G is cyclic.

3. Let A be the Abelian group (written additively) generated by x_1, x_2, x_3 satisfying the relations

$$4x_1 - 2x_2 + 4x_3 = 0$$

$$7x_1 - 8x_2 + x_3 = 0$$

$$8x_1 + x_2 + 13x_3 = 0$$

Find the order of A .

4. Show that $\mathbb{Z}[\sqrt{10}] = \mathbb{Z} + \mathbb{Z}\sqrt{10}$ is not a factorial (unique factorization) domain. Hint: show that

a) $n^2 - 10m^2 \neq 2, 3$ for all $n, m \in \mathbb{Z}$.

b) 2, 3, and $4 \pm \sqrt{10}$ are primes in $\mathbb{Z}[\sqrt{10}]$.

5. Find the Jordan canonical form of the $n \times n$ matrix $A = (a_{ij})$ with $a_{ij} = 1$ for $i \leq j$ and $a_{ij} = 0$ for $i > j$.

6. Let $K/\mathbb{Q}$ be a splitting field of $x^3 - 9x + 12$. Show that there is a single normal extension $L/\mathbb{Q}$ in K with $\mathbb{Q} \neq L \neq K$. Find $[L : \mathbb{Q}]$.

7. Show that every element of $\text{SO}_5(\mathbb{R})$ has a fixed point ($Ax = x$ for some $x \neq 0$). Recall that

$$\text{SO}_5(\mathbb{R}) = \{A \in \mathbb{R}_5 : (Ax, Ay) = (x, y), \text{ and } \det A = 1\}$$

where $(x, y) = x \cdot y = x^t y$ is the usual bilinear form (dot product) on $\mathbb{R}^5$.

Algebra General Exam, August 24, 1996

ALGEBRA GENERAL EXAMINATION

August 24, 1996

Instructions. This algebra examination is closed book. Please write out your answers carefully, being sure to explain any claims you make. Include your scratch work with what you turn in. The time allowed is three hours, and there are eight questions.

1. A Hall subgroup H of a finite group G is a subgroup whose order and index are relatively prime. Use isomorphism theorems to prove that if N is a normal subgroup of G and H is a Hall subgroup of G , then HN/N is a Hall subgroup of G/N , and $H \cap N$ is a Hall subgroup of N .
2. Let C_n denote the cyclic group of order n . Determine all pairs of positive integers a and b , with $a \leq b$, for which $C_a \times C_b$ is isomorphic to $C_{15} \times C_{18} \times C_{20}$.
3. Let M be a finitely generated R -module, R a ring. Show that any generating set of M (a subset of M that generates M as an R -module) must contain a finite generating set. Conclude that M has a minimal generating set (no proper subset generates M), and that every minimal generating set of M is finite.
4. In $\mathbb{Q}[x]$, let $f(x) = x^{m_1} + \dots + x^{m_k}$, where $m_i \equiv i - 1 \pmod{k}$. Show that $f(x)$ is divisible by $x^{k-1} + x^{k-2} + \dots + 1$.
5. Let R be a principal ideal domain. An ideal P of R is called primary if whenever $ab \in P$ and $a \notin P$, then $b^n \in P$ for some n (perhaps depending on b). Show that P is a primary ideal if and only if either $P = \{0\}$ or $P = (p^m)$ for some prime p and exponent m .
6. Let E/F be a finite Galois extension with Galois group G . The normal basis theorem states that there is an element u in E whose images under the members of G form an F -basis of E .

Prove that for any subgroup H of G , the subfield corresponding to H in the Galois correspondence is $F(u_H)$, where

$$u_H = \sum_{h \in H} h(u)$$

7. The minimal polynomial of a matrix A with real entries is $(x - 1)(x^2 + 1)^2$ and its characteristic polynomial is $(x - 1)^3(x^2 + 1)^2$.
(a) What is the rational canonical form of A ?

(b) If A is regarded as having complex entries, what is its rational canonical form?

(c) What is the Jordan normal form of A (again, with complex entries)?

8. Give an example of two finite groups whose Sylow subgroups are isomorphic for each prime, but which are themselves not isomorphic.

Algebra General Exam, January 23, 1997

ALGEBRA GENERAL EXAMINATION

January 23, 1997

This algebra examination is closed book. Please write out your answers carefully, being sure to explain any claims you make. Include your scratch work with what you turn in. The allowed time is three hours, and there are seven questions.

1. Find the rational canonical form and the Jordan normal form for the real matrix

$$\begin{pmatrix} -1 & 3 & 0 \\ 0 & 2 & 0 \\ 2 & 1 & -1 \end{pmatrix}$$

2. Let G be a finite Abelian group of order k . Use the fact that the map $g \rightarrow g^n$ (n an integer) is a homomorphism, to show that if k_n is the number of solutions of $g^n = 1$ in G , and $k^{(n)}$ is the number of n -th powers in G , then $k = k_n k^{(n)}$.
3. Let E be a separable extension of the field F , with $[E : F] = n$. Use Galois theory to find an upper bound $B(n)$ for the number of intermediate fields $K, F \subseteq K \subseteq E$, that depends only on n . You don't need to make the bound $B(n)$ very tight!
4. Give counterexamples for each of the following statements, with details. Then correct each statement by modifying the underlined part.
 - (a) If R is a commutative ring, a polynomial in $R[x]$ of degree n has at most n roots in R .
 - (b) If R is a division ring, a polynomial in $R[x]$ of degree n has at most n roots in R .
 - (c) If R is a unique factorization domain, then the greatest common divisor, d , of two members a and b of R can be written as $d = ax + by$ for some x and y in R .
5. Describe the Galois group of $x^4 - 5$ over the rationals as a group of permutations of the roots.
6. Let G be a finite group and P a p -Sylow subgroup of G for the prime p . Prove that if H is a subgroup of G , then for some $g \in G$, $H \cap gPg^{-1}$ is a p -Sylow subgroup of H .
7. Let V be a finite-dimensional vector space over a field F , and let λ and μ be two linear functionals on V . Define

$$B(x, y) = \lambda(x)\mu(y) - \lambda(y)\mu(x)$$

for x and y in V . Show that B is an alternate (alternating) form on V and determine the possible values for its rank.

Algebra General Exam, January 17, 1998

ALGEBRA GENERAL EXAMINATION

January 17, 1998

Question 1 (a) What can you say about groups G of the following orders $|G|$? (Give reasons, making free use of any theorems you know.)

(1) $|G| = 2^4 + 1 : G \cong ?$

(2) $|G| = 2^3 + 1 : G \cong ?$

(b) What can you say about simple groups G of the following orders $|G|$? (Again, explain your answer.)

(1) $|G| = 2^m + 1 : G \cong ?$

(2) $|G| = 2 \cdot 3 \cdot 5 : G \cong ?$

(3) $|G| = 2^2 \cdot 3 \cdot 5 : G \cong ?$

All your answers should be quite short.

Question 2 (a) How many non-isomorphic abelian groups are there of order 360 ?

(b) Take the free abelian group on generators x, y and z . Divide it by the relations

$$2x + 4y + 5z = 0$$

$$6x + 8y + 10z = 0$$

$$8x + 12y + 20z = 0$$

Write the resulting group as a direct sum of cyclic groups.

Question 3 Let R be a principal ideal domain. Suppose a and b are two non-zero elements in R . Show that they have a least common multiple. That means the following. We adopt the notation that $a \mid m$ means " a divides m ". You need to show that there exists an element $m \in R$, so that

3.1 $a \mid m$ and $b \mid m$.

3.2 If $a \mid x$ and $b \mid x$, then $m \mid x$.

Question 4 Let M be a module over a ring. A section $A : B$ of M is a pair of submodules A, B of M with $B \subseteq A$; a trivial section is where $B = A$. A submodule C of M covers a section $A : B$ if $(A \cap C) + B = A$, and avoids $A : B$ if $A \cap C \subseteq B$.

Algebra General Exam, August 14, 2000

General Exam - Algebra, August 14, 2000

1. Let V be a non-zero vector space over the field of reals R . Show that V cannot be presented as the union of three proper subspaces.
2. For a field K , we denote by $K(x)$ the field of rational functions in one variable x over K . Prove that $C(x)$ is not the algebraic closure of $R(x)$. (R is the real numbers and C the complex numbers.)
3. (i). Let G be a finite group of order $p^\alpha m$ where p is a prime relatively prime to m , and let G_p denote the set of elements of G whose order is p^i for some $i > 0$. Show that $|G_p| \leq [G : N] (p^\alpha - 1)$ where N is the normalizer of some Sylow p -subgroup, with equality holding iff any two distinct Sylow p -subgroups have trivial intersection.
(ii). Let G be a finite group of order $p^\alpha q^\beta$ for distinct primes p and q . Assume that G does not have elements of order pq , and that G has just 1 Sylow q -subgroup.

Show that $|G_p| = q^\beta (p^\alpha - 1)$.

Show that any two distinct Sylow p -subgroups have only the identity in common.

4. Let A denote the set of complex numbers of the form $a + b\sqrt{-6}$, with $a, b \in \mathbb{Z}$.

(i) Show that A is a subring of $\mathbb{C}$.

(ii) Let J be the ideal of A generated by 5 and $2 + \sqrt{-6}$. Show that $J \neq A$.

(iii) Show that J is prime.

(iv) Show that the ideal $5A$ is not prime.

5. Let A be a commutative ring with 1. Show that $A[x, x^{-1}]$ is Noetherian if and only if A is. (Reminder. A Noetherian ring is one in which every ideal is finitely generated. You may use any characterization of a Noetherian ring you know; you do not need to justify it.)

6. Let R be a ring with 1 and suppose that N is a non-zero left R -module. Show that N is indecomposable if and only if $\text{Hom}_R(N, N)$ does not contain an idempotent e with $e \neq 0$ and $e \neq 1$.

(Reminders. A module is indecomposable if it is not the direct sum of two non-zero submodules. An element s of a ring is idempotent if $s^2 = s$.)

7. Let K be a Galois extension of $\mathbb{Q}$, the rational numbers, with the Galois group $G(K : \mathbb{Q})$ isomorphic to A_4 . (A_4 is the group of even permutations of 4 objects.)

(i) What is the degree of K over $\mathbb{Q}$?

(ii) Show that there is a polynomial f of degree 4 so that K is isomorphic to the splitting field of f over $\mathbb{Q}$.

Let r_1, r_2, r_3, r_4 be the roots of f in K . Let $\Delta = \prod_{1 \leq i < j \leq 4} (r_i - r_j)$.

(iii) Show that Δ^2 is in $\mathbb{Q}$.

(iv) Is Δ in $\mathbb{Q}$? Why or why not?

8. A certain $\mathbb{Z}$ -module M is generated by elements a, b , and c . Furthermore $2a + 4b + 2c = 0$, $4a + 4b + 4c = 0$, and $2a + 4b + 14c = 0$. Assume all other $\mathbb{Z}$ -combinations of a, b, c which are 0 can be derived from these three. (i.e. these are the defining relations for the module.) Find the invariant factors for this module and describe M up to isomorphism as a direct sum of cyclic submodules.

9. Let X be an invertible matrix with complex entries and let $Y = X^{-1}$.

(i). Show that if λ is an eigenvalue for X then λ^{-1} is an eigenvalue for Y .

(ii). Show that if the Jordan canonical form of X has a $k \times k$ Jordan block with diagonal entries λ , then the Jordan canonical form of Y has a $k \times k$ Jordan block with diagonal entries λ^{-1} .

Proofs for the Algebra General Exam, August 21, 1999

PROOFS

Proof (1): (1) $N_M(P) = M \cap N_G(P)$ (always) $= N_G(P)$ (when $M \supseteq N_G(P)$). (2) Using Sylow on both G and M (noting that P remains a p -Sylow subgroup of M) shows the indexes of the normalizers is congruent to 1, so

$$[G : N_G(P)] \equiv 1, [M : N_G(P)] = [M : N_M(P)] \equiv 1, 1 \equiv [G : N_G(P)] = [G : M][M : N_G(P)] \equiv [G : M]1 = [G : M]$$

.

Proof (2a): Left-regular representation.

Proof (2b): G_p is in fact the set of all elements of order 1 or p [any element g of order p generates a cyclic subgroup of order p , which by uniqueness must be G_p , therefore $g \in G_p$], hence is invariant under any homomorphism.

Proof (2c): For any $s \in S$ we have $|G| > |S| \geq G \cdot s = [G : \text{Stab}(s)]$ (orbit size formula) $= |G| / |\text{Stab}(s)|$ so $|\text{Stab}(s)| > 1$. Thus $|\text{Stab}(s)|$ is a power of p , in particular (by Cauchy) has an element of order p , hence a cyclic subgroup of order p , which by uniqueness must be $G_p : G_p \subseteq \text{Stab}(s)$. Thus G_p fixes all points s , thus acts trivially on S .

Proof (3): In $A = \bigoplus Z_{n_i}$, any $x = \sum x_i$ has $x^{n_1} = \sum x_i^{n_1} = 1$ since n_1 is a multiple of n_i , and this is the smallest such power since the generator of Z_{n_1} has order exactly n_1 . If m divides n_1 then there exists $Z_m \subseteq Z_{n_1} \subseteq A$. Conversely, if A has an element x of order m then m divides the exponent n_1 (by the above).

Proof (4) The automorphisms are precisely all "linear" transformations $\varphi_{a,b}(f(x)) = f(ax + b)$ for invertible a and arbitrary b in R , with inverse $\varphi_{a^{-1}, -a^{-1}b}$. Indeed, if $\varphi(x) = f(x)$, $\varphi^{-1}(x) = g(x)$, for f, g of degrees n, m , then $x = \varphi(\varphi^{-1}(x)) = \varphi(g(x)) = g(f(x))$ shows $1 = mn$ (over a domain the degree of the product is $n + m$ and the degree of the composite is nm), so $n = m = 1$, $f(x) = ax + b$, $g(x) = cx + d$, $x = c(ax + b) + d = (ca)x + (cb + d)$ implies $ca = 1$ (so a is invertible with inverse c) and $cb + d = 0$ (so $d = -cb$).

$$\text{Proof (5): } \frac{1}{n} \sum_{i=1}^n \alpha_i = -\frac{1}{n} \left(\frac{a_{n-1}}{a_n} \right), \frac{1}{n-1} \sum_{i=1}^{n-1} \alpha'_i = -\frac{1}{n-1} \left(\frac{(n-1)a_{n-1}}{na_n} \right).$$

Proof (6) (1) if the distinct diagonal entries of the Jordan form A are λ, μ then A^2 has diagonal entries λ^2, μ^2 then we either have (1a) $\lambda = \lambda^2, \mu = \mu^2$, so $\lambda, \mu = 0, 1$ and by distinctness $\lambda = 1, \mu = 0$ and we have an idempotent $A \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, or we have (1b) $\lambda = \mu^2, \mu = \lambda^2$, so

$\lambda^4 = \lambda \neq 0, 1, \lambda^3 = 1, \lambda = \zeta$ is a primitive cube root of unity and $\mu = \lambda^2 = \lambda^{-1}$, and we have

$A \sim \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$. (2) If $A = \lambda I$ then $A^2 = \lambda^2 I$ and we must have $\lambda = \lambda^2$, so again we have

idempotents $A = I, 0$. (3) If $A \sim \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ is a 2×2 Jordan block, then $A^2 \sim \begin{pmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{pmatrix}$, so again $\lambda = \lambda^2$, $\lambda = 1, 0$; but we can't have $\lambda = 0$ (A would be nilpotent of index 2, $A^2 = 0$ nilpotent of index 1, so not similar), hence $A \sim \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $A^2 \sim \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \sim A$.

Proof (7a) Any $p(x)$ has root A consisting of n blocks B_2 strung together down the diagonal (hence the need for even size $2n$), where B_2 is the 2×2 companion matrix of $p(x)$, namely $\begin{pmatrix} -\alpha & 1 \\ -\beta & 0 \end{pmatrix}$, and Z_{n-2} the $(n-2) \times (n-2)$ zero matrix. More generally, any polynomial $p(x) = x^d + \alpha_{d-1}x^{d-1} + \dots + \alpha_1x + \alpha_0$ has root obtained by duplicating its $d \times d$ companion matrix $n \geq 1$ times to obtain a $dn \times dn$ matrix.

Proof (7b): There are infinitely many distinct conjugates of any solution A , and there is only one Jordan form for a real matrix having minimum polynomial dividing $x^2 + 1$, namely $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ with rational canonical

Proof (9): $\mathbb{Z}_p^\times$ is a cyclic group of order $p-1$ divisible by 8, so it has a unique cyclic subgroup $\langle \varepsilon \rangle$ of order 8; ε^4 is not 1, yet it is a square root of unity in the field, so it must be -1. The $(\varepsilon + \varepsilon^{-1})^2 = \varepsilon^2 + 2 + \varepsilon^{-2} = 2$ because $\varepsilon^{-2}, \varepsilon^2$ are negatives of each other: both have the same square $(-1)^{-1} = -1$, yet they are not equal (since $\varepsilon^4 \neq 1$), so their quotient squares to 1 but isn't one, and must be the only other square root of unity, namely -1.

General Exam - Algebra

General Exam - Algebra

August 14, 2000

1. Let V be a vector space over the field of reals $\mathbb{R}$. Show that V cannot be presented as the union of three proper subspaces.
2. Let K be an arbitrary field, G be a finite group of order $n > 1$. Show that the group algebra $K[G]$ has zero divisors. Does this result remain true for infinite groups?
3. Prove or disprove: A commutative local ring has only one nonzero prime ideal (by definition, a commutative ring is called local if it has only one maximal ideal).
4. For a field K , we denote by $K(x)$ the field of rational functions in one variable x over K . Prove or disprove: $\mathbb{C}(x)$ is the algebraic closure of $\mathbb{R}(x)$.
5. Let K be the splitting field of the polynomial $f(x) = x^{13} - 1$ over $\mathbf{F}_5$ (the field of 5 elements). Determine the Galois group $\text{Gal}(K/\mathbf{F}_5)$.
5. Let a and b be positive relatively prime integers. Show that any integer $N > ab$ can be written in the form $N = ax + by$ where x, y are integers ≥ 0 .
6. (a) Let G be a finite group of order $p^\alpha m$ where p is a prime relatively prime to m , and let G_p denote the set of elements of G whose order is p^i for some $i > 0$. Show that

$$|G_p| \leq [G : N] (p^\alpha - 1)$$

where N is the normalizer of some Sylow p -subgroup, with equality holding iff any two distinct Sylow p -subgroups have trivial intersection.

(b) Let G be a finite group of order $p^\alpha q^\beta$ for distinct primes p, q . Assume that G does not have elements of mixed order (i.e. the order of any element is either p^i or q^j) and has a normal Sylow q -subgroup. Show that

$$|G_p| = q^\beta (p^\alpha - 1)$$

Using (a), conclude that in this case any two distinct Sylow p -subgroups intersect trivially.

8. Determine all possibilities for the Jordan canonical form of a matrix $A \in GL_2(\mathbb{C})$ given that A is conjugate to ${}^t A^{-1}$ where t denotes the operation of taking the transpose of a matrix.

9. Let R denote the set of complex numbers of the form $a + b\sqrt{-6}$ with $a, b \in \mathbb{Z}$.

(i) Show that R is a subring of $\mathbb{C}$.

(ii) Let $\mathfrak{a}$ be the ideal of R generated by 5 and $2 + \sqrt{-6}$. Show that $\mathfrak{a}$ is proper (i.e. $\mathfrak{a} \neq R$). (Hint. Use the conjugate $2 - \sqrt{-6}$.)

(iii) Show that the ideal $\mathfrak{a}$ is prime, while the ideal $\mathfrak{b} \subset R$ generated by 5 alone is not prime. (Hint. For the first part, try to identify the quotient ring $R/\mathfrak{a}$.)

10. Let R be a unital commutative ring. From the usual theory of determinants for matrices over R , we know that if $M \simeq R^n$ is a free module we can uniquely define the determinant $\det(T)$ to any $T \in \text{End}_R(M)$. Suppose M is only a direct summand of a free module, $M \oplus M' \simeq R^n$ for some (finite) n . Define $\tilde{T} = T \oplus 1_{M'} \in \text{End}_R(M \oplus M')$ and set

$$\det_{M,M'}(T) := \det(\tilde{T})$$

Show that this is independent of the complement chosen: if also $M \oplus M'' \simeq R^m$ and $\hat{T} = T \oplus 1_{M''}$, then $\det(\hat{T}) = \det(\tilde{T})$ as elements of R . (Hint. Show that there are two

$$S_1, S_2 \in \text{End}_R(M \oplus M' \oplus M \oplus M'') \simeq \text{End}_R(R^{n+m})$$

with $\det(S_1) = \det_{M,M'}(T)$, $\det(S_2) = \det_{M,M''}(T)$, and $S_2 = PS_1P^{-1}$ for an invertible $P \in \text{End}_R(M \oplus M' \oplus M \oplus M'')$.) Conclude that there is a welldefined determinant for endomorphisms of such M 's, which agrees with the usual determinant for free modules M .

Algebra General Exam, January 17, 1998

ALGEBRA GENERAL EXAMINATION

January 17, 1998

Question 1 (a) What can you say about groups G of the following orders $|G|$? (Give reasons, making free use of any theorems you know.)

(1) $|G| = 2^4 + 1 : G \cong ?$

(2) $|G| = 2^3 + 1 : G \cong ?$

(b) What can you say about simple groups G of the following orders $|G|$? (Again, explain your answer.)

(1) $|G| = 2^m + 1 : G \cong ?$

(2) $|G| = 2.3.5 : G \cong ?$

(3) $|G| = 2^2.3.5 : G \cong ?$

All your answers should be quite short.

Question 2 (a) How many non-isomorphic abelian groups are there of order 360?

(b) Take the free abelian group on generators x, y and z . Divide it by the relations

$$2x + 4y + 5z = 0$$

$$6x + 8y + 10z = 0$$

$$8x + 12y + 20z = 0$$

Write the resulting group as a direct sum of cyclic groups.

Question 3 Let R be a principal ideal domain. Suppose a and b are two non-zero elements in R . Show that they have a least common multiple. That means the following. We adopt the notation that $a \mid m$ means " a divides m ". You need to show that there exists an element $m \in R$, so that

$$a \mid m \text{ and } b \mid m.$$

If $a \mid x$ and $b \mid x$, then $m \mid x$.

Question 4 Let M be a module over a ring. A section $A : B$ of M is a pair of submodules A, B of M with $B \subseteq A$; a trivial section is where $B = A$. A submodule C of M covers a section $A : B$ if $(A \cap C) + B = A$, and avoids $A : B$ if $A \cap C \subseteq B$.

(a) Show that C covers $A : B$ if and only if $A \subseteq B + C$, and avoids $A : B$ if and only if $A \cap (B + C) = B$.

(b) Show that every C simultaneously covers and avoids any trivial section, that any C with $C \supseteq A$ covers

$A : B$, and any C with $C \subseteq B$ avoids $A : B$.

(c) Give an example of a Z -module M and a submodule C that covers one nontrivial section $A : B$ and avoids another nontrivial section $A' : B'$ for which the two quotients A/B and A'/B' are isomorphic.

Question 5 (a) Find the Galois group of the polynomial

$$x^5 + 5x^3 - 20x + 10$$

over the rational numbers.

(b) The Galois group G over F of a polynomial $f(x) \in F[x]$ of degree 4 is known to contain a subgroup isomorphic to the dihedral group D_4 .

(1) Show that $f(x)$ is irreducible.

(2) If some root r_i of $f(x)$ lies in the subfield $F(r_j, r_k)$ generated by two other roots, show $F(r_j, r_k)$ is a splitting field for $f(x)$ and that $G = D_4$.

(3) Show that no root r_i of $f(x)$ is a linear combination of two other roots r_j, r_k .

Question 6 The finite field $GF(32)$ of 32 elements can be constructed as the extension $GF(2)(\theta)$, where θ is a root of the polynomial $x^5 + x^2 + 1$ in $GF(2)[x]$. Find the minimal polynomial of θ^3 over $GF(2)$.

Question 7 If the $n \times n$ real matrix P is the transition matrix for a regular Markov chain, then its powers

converge to a matrix $T = \lim_{k \rightarrow \infty} P^k$ of the form $\begin{bmatrix} t_1 & t_1 & \dots & t_1 \\ \dots & \dots & \dots & \dots \\ t_n & t_n & \dots & t_n \end{bmatrix}$ all of whose columns are the

same probability vector $\mathbf{t}$ (it's entries t_i are all nonnegative and sum to 1).

(a) What is the Jordan canonical form of the limit T ?

(b) What are the possible complex eigenvalues of the original P ?

(c) What are the possible Jordan canonical forms of P ?

If you cannot do the general case, do at least the 2×2 case.

Proofs for the Algebra General Exam, August 21, 1999

PROOFS

Proof (1): (1) $N_M(P) = M \cap N_G(P)$ (always) $= N_G(P)$ (when $M \supseteq N_G(P)$). (2) Using Sylow on both G and M (noting that P remains a p -Sylow subgroup of M) shows the indexes of the normalizers is congruent to 1, so

$$[G : N_G(P)] \equiv 1, [M : N_G(P)] = [M : N_M(P)] \equiv 1, 1 \equiv [G : N_G(P)] = [G : M][M : N_G(P)] \equiv [G : M]1 = [G : M]$$

.

Proof (2a): Left-regular representation.

Proof (2b): G_p is in fact the set of all elements of order 1 or p [any element g of order p generates a cyclic subgroup of order p , which by uniqueness must be G_p , therefore $g \in G_p$], hence is invariant under any homomorphism.

Proof (2c): For any $s \in S$ we have $|G| > |S| \geq G \cdot s = [G : \text{Stab}(s)]$ (orbit size formula) $= |G| / |\text{Stab}(s)|$ so $|\text{Stab}(s)| > 1$. Thus $|\text{Stab}(s)|$ is a power of p , in particular (by Cauchy) has an element of order p , hence a cyclic subgroup of order p , which by uniqueness must be $G_p : G_p \subseteq \text{Stab}(s)$. Thus G_p fixes all points s , thus acts trivially on S .

Proof (3): In $A = \bigoplus Z_{n_i}$, any $x = \sum x_i$ has $x^{n_1} = \sum x_i^{n_1} = 1$ since n_1 is a multiple of n_i , and this is the smallest such power since the generator of Z_{n_1} has order exactly n_1 . If m divides n_1 then there exists $Z_m \subseteq Z_{n_1} \subseteq A$. Conversely, if A has an element x of order m then m divides the exponent n_1 (by the above).

Proof (4) The automorphisms are precisely all "linear" transformations $\varphi_{a,b}(f(x)) = f(ax + b)$ for invertible a and arbitrary b in R , with inverse $\varphi_{a^{-1}, -a^{-1}b}$. Indeed, if $\varphi(x) = f(x)$, $\varphi^{-1}(x) = g(x)$, for f, g of degrees n, m , then $x = \varphi(\varphi^{-1}(x)) = \varphi(g(x)) = g(f(x))$ shows $1 = mn$ (over a domain the degree of the product is $n + m$ and the degree of the composite is nm), so $n = m = 1$, $f(x) = ax + b$, $g(x) = cx + d$, $x = c(ax + b) + d = (ca)x + (cb + d)$ implies $ca = 1$ (so a is invertible with inverse c) and $cb + d = 0$ (so $d = -cb$).

$$\text{Proof (5): } \frac{1}{n} \sum_{i=1}^n \alpha_i = -\frac{1}{n} \left(\frac{a_{n-1}}{a_n} \right), \frac{1}{n-1} \sum_{i=1}^{n-1} \alpha'_i = -\frac{1}{n-1} \left(\frac{(n-1)a_{n-1}}{na_n} \right).$$

Proof (6) (1) if the distinct diagonal entries of the Jordan form A are λ, μ then A^2 has diagonal entries λ^2, μ^2 then we either have (1a) $\lambda = \lambda^2, \mu = \mu^2$, so $\lambda, \mu = 0, 1$ and by distinctness $\lambda = 1, \mu = 0$ and we have an idempotent $A \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, or we have (1b) $\lambda = \mu^2, \mu = \lambda^2$, so

$\lambda^4 = \lambda \neq 0, 1, \lambda^3 = 1, \lambda = \zeta$ is a primitive cube root of unity and $\mu = \lambda^2 = \lambda^{-1}$, and we have

$A \sim \begin{pmatrix} \zeta & 0 \\ 0 & \zeta^{-1} \end{pmatrix}$. (2) If $A = \lambda I$ then $A^2 = \lambda^2 I$ and we must have $\lambda = \lambda^2$, so again we have

idempotents $A = I, 0$. (3) If $A \sim \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ is a 2×2 Jordan block, then $A^2 \sim \begin{pmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{pmatrix}$, so again $\lambda = \lambda^2$, $\lambda = 1, 0$; but we can't have $\lambda = 0$ (A would be nilpotent of index 2, $A^2 = 0$ nilpotent of index 1, so not similar), hence $A \sim \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $A^2 \sim \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \sim A$.

Proof (7a) Any $p(x)$ has root A consisting of n blocks B_2 strung together down the diagonal (hence the need for even size $2n$), where B_2 is the 2×2 companion matrix of $p(x)$, namely $\begin{pmatrix} -\alpha & 1 \\ -\beta & 0 \end{pmatrix}$, and Z_{n-2} the $(n-2) \times (n-2)$ zero matrix. More generally, any polynomial $p(x) = x^d + \alpha_{d-1}x^{d-1} + \dots + \alpha_1x + \alpha_0$ has root obtained by duplicating its $d \times d$ companion matrix $n \geq 1$ times to obtain a $dn \times dn$ matrix.

Proof (7b): There are infinitely many distinct conjugates of any solution A , and there is only one Jordan form for a real matrix having minimum polynomial dividing $x^2 + 1$, namely $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ with rational canonical form $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. [Over the complexes there would be two more conjugacy classes, corresponding to $\pm iI_2$.]

Proof (8): If the characteristic is 2 then the map $\varphi(\alpha) = \alpha^2$ is injective, hence surjective, so every element is a square (and hence trivially $\alpha = \beta^2 = \beta^2 + 0^2$ the sum of two squares). Assume the characteristic is not two, so $\varphi(\alpha) = \varphi(\beta) \Leftrightarrow (\alpha/\beta)^2 = 1 \Leftrightarrow \alpha/\beta = \pm 1$ and (except at 0) the map φ is two-to-one; if the size of F is q , then the cardinality of the set $S := \varphi(F)$ of squares is $0 + \frac{1}{2}(q-1) = \frac{q+1}{2}$ (zero together with half of the $q-1$ nonzero values). For any α the sets $\alpha - S$ and S are too big to be disjoint (then F of size q would contain $(\alpha - S) \cup S$ of size $|S| + |S| = q+1$), so they intersect in some $\alpha - \beta_1^2 = \beta_2^2$.

Proof (9): $\mathbb{Z}_p^\times$ is a cyclic group of order $p-1$ divisible by 8, so it has a unique cyclic subgroup $\langle \varepsilon \rangle$ of order 8; ε^4 is not 1, yet it is a square root of unity in the field, so it must be -1. The $(\varepsilon + \varepsilon^{-1})^2 = \varepsilon^2 + 2 + \varepsilon^{-2} = 2$ because $\varepsilon^{-2}, \varepsilon^2$ are negatives of each other: both have the same square $(-1)^{-1} = -1$, yet they are not equal (since $\varepsilon^4 \neq 1$), so their quotient squares to 1 but isn't one, and must be the only other square root of unity, namely -1.

Partial Proofs for the Algebra General Exam, August 21, 1999

PROOFS

Proof (1):

Proof (2a): Left-regular representation.

Proof (2b):

Proof (2c):

Proof (3):

Proof (4)

Proof (5):

Proof (6)

Proof (7a) Any $p(x)$ has root A consisting of n blocks B_2 strung together down the diagonal (hence the need for even size $2n$), where B_2 is the 2×2 companion matrix of $p(x)$, namely $\begin{pmatrix} -\alpha & 1 \\ -\beta & 0 \end{pmatrix}$, and Z_{n-2} the $(n-2) \times (n-2)$ zero matrix. More generally, any polynomial $p(x) = x^d + \alpha_{d-1}x^{d-1} + \dots + \alpha_1x + \alpha_0$ has root obtained by duplicating its $d \times d$ companion matrix $n \geq 1$ times to obtain a $dn \times dn$ matrix.

Proof (7b):

Proof (8):

Algebra General Exam

Algebra General Exam

August 18, 2001

- (1) If $\phi : G_1 \rightarrow G_2$ is a homomorphism of groups, and $N_1 \triangleleft G_1$, $N_2 \triangleleft G_2$ are two normal subgroups, show that the map $\bar{\phi}$ given on cosets by $\bar{\phi}(xN_1) = \phi(x)N_2$ is a well-defined homomorphism $\phi : G_1/N_1 \rightarrow G_2/N_2$ of quotient groups if and only if the original homomorphism satisfies $\phi(N_1) \subseteq N_2$.
- (2) Show that if p is a prime and H is a subgroup of G , then the number of distinct p -Sylow subgroups of H is less than or equal to the number of distinct p -Sylow subgroups of G .
- (3) How many nonisomorphic abelian groups are there of order 325 ?
- (4) If x is an element of a PID R , show that the left R -module R/Rx is irreducible as a module if and only if the element x is irreducible as an element.
- (5) How many non-similar linear transformations T can there be on an 8-dimensional real vector space V having the minimum polynomial $\mu_T(t) = (t-2)(t-3)(t-6)^3$? List their Jordan canonical forms.
- (6) Consider the transformation T on $\mathbb{R}^3$ which rotates points around the z -axis through a fixed angle θ .
- (a) Find the matrix of T with respect to the canonical basis for $\mathbb{R}^3$.
- (b) Find the characteristic polynomial of T , and factor it into irreducibles (ignore the cases when θ is an integer multiple of π).
- (c) Find the Jordan canonical form for T (you may have to pass to complex matrices).
- (7)(a) Show that for any two elements a, b in a cyclic group, at least one of the 3 elements a, b, ab is a square (i.e. is of the form x^2 for some element x in the group).
- (b) Show that the polynomial $f(x) := (x^2 - 2)(x^2 - 3)(x^2 - 6)$ has no integral roots in $\mathbb{Z}$, but for any prime $p > 0$ it has integral roots modulo p (i.e. the congruence $f(x) \equiv 0 \pmod{p}$ has an integral solution).
- (8) Find the Galois group of $f(x) := x^{13} - 1$ over the rationals $\mathbb{Q}$ (i.e. $\text{Gal}(K/\mathbb{Q})$ for K the splitting fields of $f(x)$ over $\mathbb{Q}$).

Possible alternative problems

Possible alternative problems

1. Let $p(x) \in \mathbb{R}[x]$ be a polynomial that has only positive values (i.e. $p(\alpha) \geq 0$ for any $\alpha \in \mathbb{R}$). Show that $p(x)$ can be written in the form $p(x) = u(x)^2 + v(x)^2$ for some $u(x), v(x) \in \mathbb{R}[x]$ (in other words, $p(x)$ is a sum of two squares in $\mathbb{R}[x]$).
2. Let G be a finite group of order n , and let $r \geq 1$ be an integer. Show that the map $\mu_r : G \rightarrow G, \mu_r(x) = x^r$, is surjective (i.e. for every element of G there is an r^{th} root of that element) if and only if $\text{g.c.d}(n, r) = 1$.
3. Let L/K be a field extension in characteristic zero. Suppose there exists an integer $m > 0$ such that $a^m \in K$ for all $a \in L$. Show that $L = K$. (One can give some specific value of m , for example, $m = 3$ or 5 .)
4. Let K be a field. A matrix $X \in M_n(K)$ is called nilpotent if there exists an integer $m \geq 1$ such that $X^m = 0$. Show that if $X \in M_n(K)$ is nilpotent then $X^n = 0$ (where the exponent is the same as the size of X).
5. Find an example of a finite group G such that
 - (a) $|\text{Aut}(G)| < |G|$;
 - (b) $|\text{Aut}(G)| = |G|$;
 - (c) $|\text{Aut}(G)| > |G|$.

Your work must include a rigorous proof of the fact that the required (in)equality holds.

Algebra, April 1977

Algebra, April 1977

- Let C be the field of complex numbers, and let A be an $n \times n$ matrix over C . As usual, we define $e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$, where the convergence of the series is that of C^{n^2} .
 - Use the Jordan canonical form to show that $\det e^A = e^{\text{Tr}A}$.
 - If $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, what is e^A ?
- Let k be a field of q elements where q is finite.
 - Show that $q = p^n$ is a power of a prime.
 - Show that k is the splitting field of the equation $x^q - x = 0$ and thus is the unique field of q elements.
 - The multiplicative group k^* of non-zero elements of k satisfy the equation $x^{q-1} - 1 = 0$. Why does this imply that k^* is a cyclic group of order $q - 1$?
- Use the fundamental theorem for abelian groups to show that in the ring of integers mod p^n , for p an odd prime, the multiplicative group of integers relatively prime to p has a subgroup of order p if $n > 1$.
 - Let R be a commutative ring with 1 and let M be a maximal ideal in R . Show that R/M is a field.
 - Show that if R is a Euclidean Integral Domain then every ideal generated by an irreducible element is maximal.
 - Let k be a field and $R = k[x]$ the polynomial ring in one variable over k . If $p(x) = p$ is an irreducible polynomial in $k[x]$ use 4. and (a) to show that there is a finite extension K of k in which $p(x)$ has a root.
 - If $k = Q$ is the rational field and $p(x) = x^2 + x + 1$ show that $p(x)$ is indeed irreducible over Q . What is the field generated in (b) for this example?
- Let $G = S_n$ be the symmetric group on n letters.
 - Show that the alternating group A_n on n letters is a normal subgroup of S_n .
 - Show that A_n is generated by the three-cycles $(1, 2, i), i = 3, 4, \dots, n$.
 - Show that if H is a normal subgroup of A_n for $n \geq 3$ and H contains a 3-cycle, then $H = A_n$.
- Let $k = Q$ be the rational field and let $f(x) = x^3 + 5$.
 - Show that $f(x)$ is irreducible over Q .
 - Use Galois theory to show that the Galois group of $f(x)$ cannot be of order 3.

(c) Find the Galois group of $f(x)$.

(d) Is the equation $f(x)$ solvable by radical? Give reasons for your answer.

7. Let A be a normal ($A^*A = AA^*$) $n \times n$ complex matrix. Show that there is a polynomial $f(x)$ with complex coefficients such that $A^* = f(A)$. Thus any $n \times n$ matrix B which commutes with A also commutes with A^* . (Assume the spectral theorem for A .)

8. If A and B are algebras over a field k define their tensor product $C = A \otimes_k B$ and show that C is an algebra over k .

(a) Let A and B each have units and let $Z(A) = Z_1$, $Z(B) = Z_2$, and $Z(C) = Z$ designate the centers of A , B , and C respectively. Show that $Z = Z_1 \otimes Z_2$.

Algebra, April 1977

Algebra, April 1977

1. Let C be the field of complex numbers, and let A be an $n \times n$ matrix over C . As usual, we define $e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots$, where the convergence of the series is that of C^{n^2} .
 - (a) Use the Jordan canonical form to show that $\det e^A = e^{\text{Tr}A}$.
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Algebra, November 1977

Algebra, November 1977

1. How many non-isomorphic fields are there with exactly 3 elements? Exactly 6 elements? Exactly 9 elements?
2. Describe the following tensor products of modules in more explicit form. (Z = integers).
 - (i) $Z_2 \otimes_{Z_4} Z_4$
 - (ii) $Z_2 \otimes_Z Z_3$
 - (iii) $Z_2 \otimes_{Z_3} Z$
3. Give an example of an inseparable extension of degree 7 over $\mathbb{Q}$ (the rationals).
4. Define "finite" and "algebraic" extensions of fields. Does algebraic imply finite? Does finite imply algebraic? Prove or give counterexample.
5. Give an example of a non-Noetherian commutative ring.
6. Find all eigenvectors and eigenvalues of the linear operator $D = \frac{d}{dx}$ (differentiation) on the space $V = C^\infty(\mathbb{R})$ of infinitely differentiable complex-valued functions on the real line $\mathbb{R}$. The finite-dimensional subspace V_n of complex polynomials of degree $< n$ is invariant under D ; what is the Jordan canonical form of the restriction of D to this subspace?
7. Consider groups of order 70. Are they all abelian or solvable? Can you say anything regarding their normal subgroup structure?

Algebra, November 1977

Algebra, November 1977

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6. Find all eigenvectors and eigenvalues of the linear operator $D = \frac{d}{dx}$ (differentiation) on the space $V = C^\infty(R)$ of infinitely differentiable complex-valued functions on the real line R . The finite-dimensional subspace V_n of complex polynomials of degree $< n$ is invariant under D ; what is the Jordan canonical form of the restriction of D to this subspace?
7. Consider groups of order 70. Are they all abelian or solvable? Can you say anything regarding their normal subgroup structure?

Algebra, May 1978

Algebra, May 1978

1. Give an example of a nonabelian simple group. Do not prove your assertion.
2. Let G be a finite p -group acting linearly on a finite dimensional vector space over $\mathbb{Z}_p$. Prove G has a non-zero fixed point.
3. Describe the semisimple 4 -dimensional algebras over the reals $\mathbb{R}$. (No proofs required.)
4. Describe the modules over the ring $M_n(F)$ of $n \times n$ matrices over a field F .
5. Is the ring $C[0, 1]$ of all real-valued continuous functions on the interval $[0, 1]$ Noetherian (i.e., does it satisfy the ascending chain condition on ideals)? Justify your assertion.
6. Give an example of a torsion free abelian group which is not free. Can you give an example which is finitely generated?
7. Find the inverse of the matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

8. (a) What is the Jordan canonical form of the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

- (b) Give the eigenvalues, characteristic and minimum polynomials for this matrix.
9. Show that any two eigenvectors with distinct eigenvalues of the matrix

$$\begin{pmatrix} 1 & 9 & 7 & 8 \\ 9 & 0 & 0 & 0 \\ 7 & 0 & 1 & 0 \\ 8 & 0 & 0 & 0 \end{pmatrix}$$

are orthogonal.

10. Give a polynomial whose splitting field is a field with 9 elements. Can this be done in a similar way for a field of 18 elements?

11. Say what you can about the intermediate subfields of a Galois extension E/F with Galois group the symmetric group S_3 .

12. Fill in the blank:

$$\mathbb{Z}_6 \otimes_{\mathbb{Z}} \mathbb{Z}_4 \cong$$

$$\mathbb{Z}_6 \otimes_{\mathbb{Z}} \mathbb{Z} \cong$$

$$\mathbb{Z}_6 \otimes_{\mathbb{Z}} \mathbb{Q} \cong$$

$$\text{Hom}(\mathbb{Z}_6, \mathbb{Z}_{14}) \cong$$

13. Define "projective module." Is there a dual notion?

Algebra, May 1978

Algebra, May 1978

1. Give an example of a nonabelian simple group. Do not prove your assertion.
2. Let G be a finite p -group acting linearly on a finite dimensional vector space over $\mathbb{Z}_p$. Prove G has a non-zero fixed point.
3. Describe the semisimple 4-dimensional algebras over the reals $\mathbb{R}$. (No proofs required.)
4. Describe the modules over the ring $M_n(F)$ of $n \times n$ matrices over a field F .
5. Is the ring $C[0, 1]$ of all real-valued continuous functions on the interval $[0, 1]$ Noetherian (i.e., does it satisfy the ascending chain condition on ideals)? Justify your assertion.
6. Give an example of a torsion free abelian group which is not free. Can you give an example which is finitely generated?
7. Find the inverse of the matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

8. (a) What is the Jordan canonical form of the matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

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13. Define "projective module." Is there a dual notion?

Algebra, September 1978

Algebra, September 1978

1. Prove that the set of rational 3×3 matrices which commute with the matrix

$$\begin{bmatrix} 0 & 0 & 6 \\ 1 & 0 & 4 \\ 0 & 1 & 2 \end{bmatrix}$$

is a field.

2. Let V be a 12-dimensional vector space over the field $GF(3)$. Let T be a linear transformation whose characteristic polynomial factors into linear factors over $GF(3)$. Find all possible Jordan normal forms given the table:

Transformation	Rank
T	10
T^2	9
T^3	9
$T - 1$	12
$T - 2$	9
$(T - 2)^2$	7
$(T - 2)^3$	6

3. A certain group of order 60 is known to have exactly four members of order 5. Explain why they, along with the identity, form a normal subgroup.
4. (a) Show that the Galois group over the rationals of every irreducible 5th degree polynomial contains an element of order 5 .
- (b) Show that the Galois group over the rationals of $x^4 + 1$ does not contain an element of order 4 .
- (c) Show that the Galois group over the rationals of $x^4 + x^3 + 1$ contains an element of order 4.

5. Let M be a module over an integral domain A with quotient field K . Show that M is torsion free if and only if $0 \rightarrow M \rightarrow M \otimes_A K$ is exact.
6. A local ring is defined to be a commutative ring with unit having a unique maximal ideal.
- (a) Prove a commutative ring, R , is local if and only if the non-units form an ideal.
 - (b) A domain, V , is called a valuation domain if for $a, b \in V$ either a divides b or b divides a . Show that a valuation domain is a local ring.
 - (c) Prove that a local ring has no idempotents (other than 1).
7. Let A be a simple Artinian ring, and let N be a minimal right ideal.
- (a) Viewing N as a right A -module, show that the ring $D = \text{End}_A(N)$ of A -linear endomorphisms of N is a division ring.
 - (b) Prove $AN = A$.
 - (c) Let k be minimal such that there exist $m_1, \dots, m_k$ in N and $a_1, \dots, a_k$ in A with $1 = a_1 m_1 + \dots + a_k m_k$. Prove that the map

$$\underbrace{N \oplus \dots \oplus N}_{k\text{-copies}} \rightarrow A$$

given by $(x_1, \dots, x_k) \rightarrow a_1 x_1 + \dots + a_k x_k$ is an isomorphism of right A -modules.

- (d) Deduce $A \cong \text{End}_A(N^k) \simeq M_k(D)$, the algebra of $k \times k$ matrices over D . What famous theorem is this?

Algebra, September 1978

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Algebra, May 1980

Algebra, May 1980

1. Describe all abelian groups of order 375. Be sure that you have at least listed each possibility up to isomorphism in some fashion.
2. A famous theorem of Hilbert states that $k[x_1, \dots, x_n]$ is a Noetherian ring when k is a field. Prove in detail that every commutative k -algebra, A , which can be generated as a k -algebra by a finite number of elements is also Noetherian, that is, satisfies the ascending chain condition.
3. Give the eigenvalues of the following matrices. What are their Jordan normal forms?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

4. Let $Q(X)$ be a quadratic form for $X = (x_1, x_2)$ in $\mathbb{R}^2$ given by

$$Q(X) = ax_1^2 + bx_1x_2 + cx_2^2.$$

Recall that the bilinear form $B(U, V)$ associated with a quadratic form $Q(X)$ is given by

$$B(U, V) = Q(U + V) - Q(U) - Q(V)$$

- (a) What is the associated (symmetric) bilinear form for $Q(X)$?
 - (b) Show that the associated (symmetric) bilinear form for $Q(X)$ is non-degenerate if and only if $b^2 - 4ac \neq 0$.
5. Let A be a finite-dimensional algebra over the field, $\mathbb{C}$, of complex numbers. What can you say about A in the following cases?
 - (a) A is simple.
 - (b) A is semi-simple.
 - (c) Let $T : V \rightarrow V$ be a linear transformation on the finite-dimensional vector space over $\mathbb{C}$. Let $A = \mathbb{C}[T]$ be the algebra of polynomials in T over $\mathbb{C}$. When is A a simple algebra?
 - (d) When is the algebra of part (c) semi-simple and what can you say about T in this case?

Note: In parts (c) and (d), your answer should specify conditions on the transformation T .

6. Let $F = Q(\zeta)$ where ζ is a primitive n^{th} root of 1. Let a be a non-zero element of F which is not an n^{th} root in F . Show that the galois group of the field $K = F(\sqrt[n]{a})$ over F is abelian.

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Algebra, September 1980

Algebra, September 1980

1. Describe all abelian groups of order 80. Be sure that you have listed each possibility once and only once up to isomorphism.
2. A field is a trivial example of a principal ideal domain (P.I.D.). All abelian groups are modules for another well-known P.I.D., namely the ring _____. Give a third example of a P.I.D. Show that every homomorphic image of a P.I.D. is a principal ideal ring.
3. Suppose A is a 4×4 matrix with complex entries with $p(A) = 0$ for $p(t) = t^2 - 7t + 10$ and $\text{trace } A = 11$. Knowing $\text{trace } A = 11$, it is relatively easy to find all possible Jordan canonical forms for A . Find these forms and briefly explain your reasoning.
What is the determinant of A ?
4. Let V be a finite-dimensional vector space over a field k . The dual, V^* , of V is defined as the collection of all linear maps $V \rightarrow k$ (often called "linear functionals") with the vector space operations defined in the obvious way. If $v_1, \dots, v_n$ is a basis for V , define $v_1^*, \dots, v_n^*$ in V^* as the unique linear maps $V \rightarrow k$ satisfying

$$v_i^*(v_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

- (a) Why, exactly, do there exist linear maps satisfying this condition?
(b) Show that $v_1^*, \dots, v_n^*$ is a basis for V^* (often called the "dual basis" associated with $v_1, \dots, v_n$).
5. (a) Explain what it means to say that a sequence

$$0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$$

of vector spaces and linear maps is exact.

What does it mean to say that the sequence

$$0 \rightarrow V_1 \rightarrow V_2 \rightarrow \dots \rightarrow V_n \rightarrow 0$$

of vector spaces and linear maps is exact?

- (b) If the sequence $(*)$ is exact and each V_i is of finite dimension, m_i , show that

$$m_1 - m_2 + m_3 - \dots + (-1)^{n+1} m_n = 0$$

6. Let A be a finite-dimensional algebra over the field $\mathbb{C}$ of complex numbers. What can you say about A in the following cases?
- (a) A is simple.
 - (b) A is semi-simple.

Let $T : V \rightarrow V$ be a linear transformation on the finite-dimensional vector space, V , over $\mathbb{C}$. Let $A = \mathbb{C}[T]$ be the algebra of polynomials in T over $\mathbb{C}$. Under these hypotheses:

- (c) When is A a simple algebra?
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7. Let $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. It may be shown, and you may assume, that the Galois group is of order 4 and isomorphic to the "Klein 4-group" (the direct product of a cyclic group of order 2 with itself). Describe all the subfields of F and justify your description.

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ALGEBRA GENERAL EXAMINATION - FALL 1987

ALGEBRA GENERAL EXAMINATION - FALL 1987

1. Show that a group of order 160 having two subgroups of order 80 also has a normal subgroup of order 5.
2. Prove or disprove: If R is a principal ideal domain so is $R[x]$.
3. A complex number satisfying a monic polynomial with integer coefficients is called an algebraic integer. Show that the complex number α is an algebraic integer if and only if $\mathbb{Z}[\alpha]$ is finitely generated as a $\mathbb{Z}$ -module.

4. Use the properties of tensor products to show $z_2 \otimes z_3 = 0$ and

$$z_2 \otimes z_2 \cong z_2.$$

5. Determine whether the real quadratic forms $Q_i(x) = xA_ix^t$ with

$$A_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ and } A_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ are equivalent.}$$

6. Find the rational canonical form over the real numbers of the linear transformation whose Jordan canonical form is

$$\begin{bmatrix} 2 & 1 & & & & & & & \\ 0 & 2 & & & & & \dots & & \\ & & 2 & 2 & 1 & & & & \\ & & 0 & 2 & & & & & \\ & & & & i & 1 & & & \\ & & & & 0 & i & & & \\ & & & & & & & -i & \\ & & & & & & & 0 & \\ & & & & & & & & -i \end{bmatrix}.$$

7. If K is a finite field with 9 elements, determine whether $x^2 + x - 1$ has a root in K .
8. Let K/Q be a splitting field of an irreducible polynomial $f(x)$ of degree 8. Suppose α is a root of f and that f factors over $\mathbb{Q}(\alpha)$ into 2 linear factors and 3 quadratic factors. Find the possible orders of the Galois group $\text{Gal}(K/Q)$. Show f is solvable by radicals.

Saturday August 21, 1988

9:00-12:00 AM

CLOSED BOOK. USE YOUR OWN PAPER. WRITE FINAL ANSWERS
NEATLY AND LOGICALLY (turn in scratchwork too).

PROBLEM 1.

PROVE that the ring $F[X]$ of polynomials in one variable over a field F is a principal ideal domain.

Algebra General Exam, January 1992

Algebra General Exam January 1992

Closed book. Write your final answers neatly (turn in your scratch work). Three hour time limit.

χ Prove: An infinite abelian group is cyclic if and only if every subgroup other than $\{1\}$ has finite index.

8. Show that a polynomial of degree n over a field F has at most n roots in F .

3. For φ the rationals and ω a primitive cube root of 1, determine whether $\varphi\left(\omega^3\sqrt{2}\right)$ is a Galois extension of Q . Give reasons.

4. Prove: If a finite group G has a normal p -Sylow subgroup S_p , then $\varphi(S_p) \subset S_p$ for every endomorphism φ of G .

5. A matrix E is idempotent if $E^2 = E$. Show that two idempotent matrices over a field are similar if and only if they have the same rank.

4. Prove: An integral domain has the descending chain condition on ideals if and only if it is a field.

7. Apply the Jordan-Hölder-Schreier Theorem on composition series to a finite cyclic group of order n to prove that n has a unique factorization as a product of primes.

Algebra General Exam May 1992

Algebra General Exam May 1992

Closed book. Write your final answers neatly (turn in your scratch work). Three hour time limit.

4. Show that an abelian group has a composition series if and only if it is finite.
2. Show that if $a^n = b^n$ and $a^m = b^m$ for m and n relatively prime positive integers integral domain and a and b in a commutative , then $a = b$.
3. Let V be the subgroup $\{1, (12)(34), (13)(24), (14)(23)\}$ of the symmetric group S_4 on 4 symbols. Prove that V is a normal subgroup of S_4 and that $S_4/V \cong S_3$.
4. Give the Jordan normal forms for all 6×6 matrices, over the complex numbers, whose minimal polynomials are $(x - 1)^2(x - 2)$.
5. The Euler φ -function on the natural numbers is given by: $\varphi(n) =$ the number of integers x with $1 \leq x \leq n$ and x relatively prime to n . It is a fact that $n = \sum_{d|n} \varphi(d)$. Explain this in terms of the subgroup structure of the cyclic group of order n .
6. Show that $f(x, y) = x + x^3y + y^8 + x^7y^5 + x^2y^4$ is irreducible over the rational field.
7. Show that if a 2×2 real symmetric matrix A has $\lim_{n \rightarrow \infty} \text{trace}(\hat{A}) = 0$, then $\lim_{n \rightarrow \infty} A^n = 0$. Does this hold for $n \times n$ real symmetric matrices?

Algebra General Examination September 4,1993

Algebra General Examination September 4,1993

This is a closed book exam. Please write your answers neatly and turn in your scratch work. The allowed time is three hours. There are 7 problems.

1. A real square matrix T is called tripotent if $T^3 = T$. Show that two tripotent matrices of the same size are similar if and only if they have the same rank and the same trace.
2. Let R be a commutative ring and let P be a prime ideal of R . Show that if P is not maximal, then P has infinitely many cosets in R .
3. Let N be a normal subgroup of the finite group G_5 and let P be a p -Sylow subgroup of M for some prime p . By considering conjugates of p , prove that if The normalizer of P is H , then $G = HN$.
4. Prove that it can never be the case that the additive group of a field is isomorphic to the multiplicative group of nonzero elements. [Hint: consider orders of elements in the two groups.]
5. a) Let D be a Euclidean domain, with Euclidean function δ (but do not assume $\delta(ab) \geq \delta(a)$). Let $D_n = \{a \in D \mid a \neq 0 \text{ and } \delta(a) \geq n\}$. Show that
 1. if $b \in D_0$ and there exists an $a \in D$ such that $a + Db \subseteq D_n$, then $b \in D_{n+1}$;
 2. $\cap D_n = \emptyset$.b) Conversely, show that if an integral domain D has a chain of subsets $D - \{0\} = D_0 \supseteq D_1 \supseteq D_2 \supseteq \dots$ satisfying properties 1) and 2), then D is Euclidean.
6. Let $f(x) = x^5 - 6x + 3$.
 - a) Prove that $f(x)$ is irreducible over the rational field, $\mathbb{Q}$.
 - b) Prove that $f(x)$ has exactly three real roots.
 - c) Let F be the splitting field of $f(x)$ over $\mathbb{Q}$. If E is a real subfield of F , prove that the Galois group, $g(F/E)$, of F over E contains a transposition (in the action on the roots of $f(x)$).
 - d) $\mathcal{G}(F/\mathbb{Q})$ is isomorphic to S_5 . Assuming that, prove that if $r \in \mathbb{Q}$ is positive, but $\sqrt{r} \notin \mathbb{Q}$, then $\sqrt{r} \notin F$.
7. In a well-known puzzle (figure a) one can move the tiles, numbered 1 to 15, so that at the end of a sequence of moves, the empty square is again at the lower right (figure b). The numbers 1 to 15 have been permuted by $\pi \in S_{15}$. Show That $\pi \in A_{15}$. (An allowable move consists of sliding a tile, adjacent to the empty square, horizontally or vertically, into the position

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	—

$\pi(1)$	$\pi(2)$	$\pi(3)$	$\pi(4)$
$\pi(5)$	$\pi(6)$	$\pi(7)$	$\pi(8)$
$\pi(9)$	$\pi(1)$	$\pi(1)$	$\pi(2)$
$\pi(3)$	$\pi(4)$	$\pi(15)$	—
(b)		—	—

of the empty square.)

ALGEBRA GENERAL EXAMINATION January 15, 1994

ALGEBRA GENERAL EXAMINATION January 15, 1994

The examination is closed book. Please write out your answers neatly and include your scratch work. The allowed time is three hours; there are eight problems.

Find all the possible rational canonical forms for a 5 by 5 matrix with rational entries whose minimal polynomial is $(x - 2)^2(x + 3)$. Identify a pair having the same characteristic polynomial.

2. Prove that the Galois group of $x^4 - 2$ over the rational field is isomorphic to the dihedral group of order 8.

3. Prove that $y^3 + x^2y^2 + x^3y + x$ is irreducible in $\mathbb{Z}[x, y]$.

4. Let G and H be finite groups and let $f : G \rightarrow H$ be a surjective homomorphism. Let p be a prime.

(a) Prove that if P is a p -Sylow subgroup of G , then $f(P)$ is a p -Sylow subgroup of H .

(b) Prove that every p -Sylow subgroup of H has the form $f(P)$ for some p -Sylow subgroup P of G .

5. Give an example (and a proof that it works) of a ring R without identity and an ideal in the direct sum $R \oplus R$ that does not have the form $I_1 \oplus I_2$, where I_1 and I_2 are ideals in R .

6. (a) Explain why the inner automorphism group of the alternating group A_n is isomorphic to A_n for $n \geq 4$.

(b) Prove that for $n \geq 3$, A_n has outer automorphisms.

7. In a field, it is true that if $t^m = 1$ and $t^n = 1$ for two relatively prime exponents m and n , then $t = 1$.

Explain how this implies the following result for an integral domain: if $a^m = b^m$ and $a^n = b^n$ for two relatively prime positive exponents m and n , then $a = b$.

8. Let R be a commutative ring with identity and let D be the set of zero divisors of R , including 0. Let $\mathcal{d}$ be the set of ideals of R that are contained in D .

(a) Show that if M is a maximal member of $\mathcal{d}$, then M is a prime ideal.

(b) Use Zorn's lemma to show that each member of $\mathcal{d}$ is contained in a maximal member of $\mathcal{d}$.

(c) Conclude that D is a union of prime ideals.

August 27, 1994

August 27, 1994

The examination is closed book. Please write out your answers neatly and include your scratch work. The allowed time is three hours and there are eight problems.

1. Let P be a p -Sylow subgroup of a finite group G , p a prime dividing the order of G .
 - (a) Prove that P consists of all the p -torsion elements of the normalizer $N_G(P)$, that is, all elements of $N_G(P)$ whose order is a power of p . [Hint: apply Sylow theorems to $N_G(P)$.]
 - (b) Prove that P is a characteristic subgroup of $N_G(P)$.
 - (c) Prove that $N_G(N_G(P)) = N_G(P)$.
2. If G is a finite Abelian group of order n , show that G has a subgroup of order d for each divisor d of n . Show that this need not be true if G is not Abelian.
3. Describe the subring $\mathbf{R}[x^2, x^3]$ of the polynomial ring $\mathbf{R}[x]$; that is, specify what polynomials belong to it.

4(a). Find the Jordan form of

$$A = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 2 & -1 & 3 \end{pmatrix}$$

over $\mathbf{C}$.

- 4(b). If $A \in M_n(\mathbf{C})$ has finite order d , so that d is the smallest positive exponent for which $A^d = I$, describe the Jordan form of A over $\mathbf{C}$.
5. If $f(x, y)$ is an alternating bilinear form on a 25-dimensional real vector space and A is its matrix with respect to some basis, show that $\det(A)^{25} = 0$.
6. Show that $F = \mathbf{Z}_2[x] / (x^4 + x + 1)$ is a field of characteristic 2 containing a primitive 15-th root of unity. Exhibit such a root.
7. Let ζ_n be a primitive n -th root of unity in $\mathbf{C}$ for $n > 2$.
 - (a) Show that the fixed field of $\mathbf{Q}(\zeta_n)$ under complex conjugation is $\mathbf{Q}(\zeta_n + \bar{\zeta}_n) = \mathbf{Q}(\zeta_n) \cap \mathbf{R}$. [Hint: write $\bar{\zeta}_n$ as a power of ζ_n and find a polynomial of low degree satisfied by ζ_n over $\mathbf{Q}(\zeta_n + \bar{\zeta}_n)$.]
 - (b) For $n = 7$, find the Galois group of $\mathbf{Q}(\zeta_7 + \bar{\zeta}_7)$. Then find all subfields of $\mathbf{Q}(\zeta_7)$ and their Galois groups.
8. If the annihilator of a left R -module M is $\text{Ann}_R(M) = \{a \in R \mid aM = 0\}$, show that for submodules M_1, M_2 of M , we have

$$\text{Ann}_R(M_1 + M_2) = \text{Ann}_R(M_1) \cap \text{Ann}_R(M_2).$$

Show furthermore that we have

$$\text{Ann}_R(M_1) + \text{Ann}_R(M_2) \subset \text{Ann}_R(M_1 \cap M_2);$$

but show that this inclusion could be strict.

Algebra General Examination - Saturday, August 23, 1997; 9-12

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This examination is closed book. Please explain yourself carefully and clearly; avoid words like "obviously"! Save your scratch-work and turn it in along with the examination. Allow enough time to polish your answers. There are eight problems.

1. Let p be a prime number, and let K be a field containing p distinct p -th roots of unity. Let L/K be a Galois extension for which $[L : K] = p$.
 - a) Prove that the Galois group $\text{Gal}(L/K)$ is cyclic of order p .
 - b) Let σ be a generator of $\text{Gal}(L/K)$. Then σ is a K -linear transformation on the vector space L . What is the Jordan canonical form of σ ?
 - c) Prove that there is an element u in $L - K$ with u^p in K .
2.
 - a) Let G be a finite group with exactly two conjugacy classes of elements. Prove that $|G| = 2$, where $|G|$ is the order of G .
 - b) Suppose that G has exactly three conjugacy classes of elements. Show that $|G|$ involves at most two primes.
 - c) There are, in fact, only two finite groups with exactly three conjugacy classes of elements. Can you guess which ones they are?
3. Let $K(x)$ be the field of rational functions over the field K . Prove Lüroth's theorem: for any nonconstant function, f (that is, $f \in K(x) - K$), the degree $[K(x) : K(f)]$ is finite. Can you describe $[K(x) : K(f)]$ in terms of f ?
4. Let $f(x)$ be a monic polynomial in $\mathbb{Z}[x]$. Let p be a prime and let $\bar{f}(x)$ be the image of $f(x)$ in $(\mathbb{Z}/p\mathbb{Z})[x]$ obtained by reducing the coefficients of $f(x)$ modulo p . A theorem of Kronecker says that if $\bar{f}(x)$ has distinct roots in its splitting field, and if $\bar{\sigma}$ is a member of the Galois group of $\bar{f}(x)$ over $\mathbb{Z}/p\mathbb{Z}$, then there is an element σ of the Galois group of $f(x)$ over $\mathbb{Q}$ having the same cycle structure as a permutation of the roots of $f(x)$ as $\bar{\sigma}$ does as a permutation of the roots of $\bar{f}(x)$.
 - a) Use this theorem (with a couple of choices of p) to show that the Galois group of $x^4 + 4x^2 + x + 3$ is isomorphic to S_4 .
 - b) Apply the theorem to show that if $f(x)$ has even degree and the discriminant of $f(x)$ is a perfect square in $\mathbb{Z}$, then $\bar{f}(x)$ is reducible for each prime p .

5. If g is a member of a group G , then g is called a nongenerator of G if whenever G is generated by a subset containing g , it is also generated by the subset with g removed. It is a fact that the set of all nongenerators of G forms a subgroup; that subgroup is called the Frattini subgroup of G .

Suppose that G is a finite p -group, p a prime, and F is the Frattini subgroup of G . Show that $g \in F$ if and only if g is in every subgroup of index p of G . (You may need the fact that if H is a proper subgroup of G , then H is a proper subgroup of its normalizer in G .) Conclude that G/F is an Abelian group of exponent p .

6. Let R be an integral domain and let $N : R - \{0\} \rightarrow \{n > 0 \mid n \in \mathbb{Z}\}$ be a function for which $N(1) = 1$ and $N(xy) = N(x)N(y)$, for all x, y in $R - \{0\}$.

a) Let K be the field of fractions of R . Show that N can be extended in a unique way to a function from $K - \{0\}$ into $\mathbb{Q}$ for which $N(xy) = N(x)N(y)$ still holds.

b) Show that R is a Euclidean domain (under N) if and only if for each $x \in K - \{0\}$ there is an element $r \in R$ for which $N(x - r) < 1$.

7. Let M be the free module over $\mathbb{Q}[x]$ with basis v_1, v_2, v_3 . A certain submodule N of M has basis w_1, w_2, w_3 , with

$$\begin{aligned} w_1 &= (x^3 - x^2 - x + 1)v_1 + (2x^2 + 2x)v_2 + (x^3 + x^2)v_3 \\ w_2 &= (2x^3 - 3x^2 - 3x + 2)v_1 + (5x^2 + 5x)v_2 + (2x^3 + 2x^2)v_3 \\ w_3 &= (x^3 - x)v_1 + (x^2 + x)v_2 + (x^3 + x^2)v_3 \end{aligned}$$

a) There is a theorem which implies that M has a basis b_1, b_2, b_3 for which d_1b_1, d_2b_2, d_3b_3 are a basis for N for some d_1, d_2, d_3 in $\mathbb{Q}[x]$ with d_1 dividing d_2 and d_2 dividing d_3 . Write a carefully worded statement of such a theorem, giving all pertinent hypotheses and all appropriate conclusions.

b) Find values for d_1, d_2, d_3 for the example given here.

8. a) Suppose R is a commutative ring with identity and let N be the set of all nilpotent elements of R . Show that N is an ideal; it is called the nil radical of R .

b) The intersection of all maximal ideals of R is called the Jacobson radical of R . If J is the Jacobson radical of R , show that $N \subseteq J$.

c) Show that $x \in J$ if and only if $1 + rx$ is a unit of R . [Error here: the problem should have said: $1 + rx$ is a unit for all r in R .]

d) Give an example for which N and J are different.

Algebra General Exam August 22, 1998

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(1)[20] Let G be a finite group of order $3 \cdot 5 \cdot 17$.

(a) [5] Show that the Sylow 17-subgroup is normal.

(b) [5] If there exists an element of order 15 in G , show that the Sylow 3- and 5-subgroups are also normal. [Hint: show they are properly contained in their normalizers.]

(c)[5] If all Sylow subgroups of a finite group G are normal and abelian, show that G itself is abelian.

(d)[5] If G is a finite abelian group of order $p \cdot q \cdot r$ for distinct primes p, q, r , show that G is cyclic.

(2)[15] Let $GL_2(p)$ for a prime p denote the group of invertible 2×2 matrices over the finite field F_p of p elements.

(a) [5] Find the order n of the group $GL_2(p)$.

(b) [5] For λ in F_p , find the order m of the subgroup

$$G_\lambda = \left\{ A \in GL_2(p) \mid A \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} A^{-1} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \right\}$$

(c)[5] Find how many 2×2 matrices over F_p are similar to the matrix $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. [Hint: express it in terms of m and n .]

(3)[10] For an abelian group A , the dual group A^* is defined to be $\text{Hom}(A, U)$ where the circle group U is the multiplicative group of complex numbers of modulus 1 (the unit circle in the complex plane). Here $\text{Hom}(A, B)$ denotes the abelian group of homomorphisms of A into B (under

$(f + g)(a) = f(a) + g(a)$); you may use the additivity property

$\text{Hom}(A_1 \oplus A_2, B) \cong \text{Hom}(A_1, B) \oplus \text{Hom}(A_2, B)$ and the isomorphism property that if $A \cong A', B \cong B'$ then $\text{Hom}(A, B) \cong \text{Hom}(A', B')$.

(a) [5] Prove that Z_n^* is cyclic of order n .

(b)[5] Prove that A^* is isomorphic to A for any finite abelian group A .

(4)[10] Let a, b, c be distinct elements of an integral domain D . Show that there are unique elements x, y, z in D such that

$$\begin{aligned} x + y + z &= 0 \\ ax + by + cz &= 0 \end{aligned}$$

for all elements a, b of R . Show that if D is a derivation, and in addition $D^2 = 0$ and R has no 2-torsion ($2a = 0$ implies $a = 0$), then the "exponential map" $Id + D$ is an automorphism of R .

(6)[15] Let V be an n -dimensional vector space over the complex numbers, and let T be a linear transformation from V to itself whose minimum polynomial $\mu(x)$ has degree 2.

(a) [5] Find all possible Jordan Canonical Forms for T . [Hint: consider the possible factorizations of $\mu(x)$ over K .]

(b)[5] Show that V is a direct sum of T -invariant subspaces, each of which has dimension less than or equal to 2.

(c) [5] Show that T has an eigenvalue λ such that the λ -eigenspace (the set of all eigenvectors for the eigenvalue λ , together with the zero vector) has dimension at least $n/2$.

(7)[10] Suppose that K is a finite Galois extension of the rational field Q which contains $\sqrt{3}$ and has cyclic Galois group $\text{Gal}(K/Q)$. Show that $L = Q(\sqrt{3})$ is the only quadratic extension of Q contained in K .

(8)[10] (a)[5] If F denotes a field and Z the ring of integers, decide which of the following rings are PIDs and which are UFDs (no proofs are necessary):

$$F, Z, Z[x], F[x, y], F[[x]] \text{ (formal power series).}$$

(b)[5] The Krull dimension of a commutative ring R is the longest chain of prime ideals properly contained in R , i.e. the largest integer n such that there exists a chain $P_0 < P_1 < \dots < P_n < R$ ($P_0 = 0$ allowed if prime) of prime ideals P_i in R . If R is a PID, find its Krull dimension.

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$$x + y + z = 0$$

$$ax + by + cz = 0$$

$$a^2x + b^2y + c^2z = 0$$

(5)[10] A derivation D of a ring R is a map of R into itself such that

$$\begin{aligned}D(a + b) &= D(a) + D(b) \\D(ab) &= D(a)b + aD(b)\end{aligned}$$

for all elements a, b of R . Show that if D is a derivation, and in addition $D^2 = 0$ and R has no 2-torsion ($2a = 0$ implies $a = 0$), then the "exponential map" $\exp d + D$ is an automorphism of R .

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(1)[10] Let P be a p -Sylow subgroup of a finite group G for some prime p dividing the group order. We know by Sylow theory that the index of the normalizer of P is congruent to 1 modulo p ,

$$[G : N_G(P)] \equiv 1 \pmod{p}$$

Show that the same holds for any larger subgroup M : if $G \supseteq M \supseteq N_G(P)$ then

$$[G : M] \equiv 1 \pmod{p}$$

[Hint: show $N_M(P) = N_G(P)$.]

(2)[20] (a)[5] Show that any finite group G has a faithful action on some set S of cardinality $|S| = |G|$.

From now on (in parts b,c) assume that G is a finite p -group for a prime p , having the property that it has a unique subgroup G_p of order p . [The quaternion group is such a group, with $p = 2$ and $G_2 = \{1, -1\}$.]

(b)[5] Show that G_p is invariant under all endomorphisms of G , $f(G_p) \subseteq G_p$ for all homomorphisms $f : G \rightarrow G$.

(c)[10] Show that G_p "needs room" in order to act: whenever G acts on a finite set S of size $|S| < |G|$, the subgroup G_p acts trivially. Conclude that G can only act faithfully on sets of size $\geq |G|$.

(3)[10] The exponent e of a group G is defined as the smallest positive integer k such that $x^k = 1$ for all $x \in G$; for an abelian group, in additive notation this is the smallest k such that $kx = 0$ for all elements. If $n_1, n_2, \dots, n_r$ are the invariant factors of a finite abelian group A (so $n_r \mid n_{r-1} \mid \dots \mid n_2 \mid n_1$), prove that A has exponent $e = n_1$, and has an element of order precisely e . Conclude that A has an element of order m iff m divides the largest invariant factor n_1 .

(4)[10] If R is a (commutative, unital) integral domain, find all R -linear automorphisms of the polynomial ring $R[x]$ (all ring automorphisms φ with $\varphi(a) = a$ for all constant polynomials $a \in R$). [If you can't do the general case, do the case when $R = F$ is a field.]

(5)[10] Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ ($a_0 \neq 0$) be a complex polynomial of degree $n > 1$, and f' its derivative. Let $\alpha_1, \dots, \alpha_n$ be the n roots of f and $\alpha'_1, \dots, \alpha'_{n-1}$ the $n - 1$ roots of the derivative (listing each root as many times as its multiplicity). Show that the average of the roots $\alpha_1, \dots, \alpha_n$ of f equals the average of the roots $\alpha'_1, \dots, \alpha'_{n-1}$ of f' . [Hint: Use the relations between the coefficients of a polynomial and the roots of that polynomial.]

(6)[10] Describe (in terms of Jordan form) all 2×2 complex matrices which are similar to their square. [Hint: there are more matrices, Horatio, than are dreamt of in your philosophy!]

- (7)[10] (a)[5] Show that the ring of $2n \times 2n$ matrices $M_{2n}(F)$ over a field F for $n \geq 1$ is "algebraically closed" with respect to polynomials of degree 2, in the sense that every polynomial $p(x) = x^2 + \alpha x + \beta \in F[x]$ of degree 2 has a "root" $A \in M_{2n}(F)$ (in the sense that $p(A)$ is the zero matrix). Can you generalize this to polynomials of degree d ?
- (b)[5] How many real 2×2 matrices $A \in M_2(\mathbb{R})$ are roots of the polynomial $p(x) = x^2 + 1$?
- (8)[10] If F is a finite field show that every element $\alpha \in F$ is the sum $\alpha = \beta_1^2 + \beta_2^2$ of two squares (for some $\beta_1, \beta_2 \in F$).
- (9)[10] If p is a prime number congruent to 1 mod 8, show that 2 is a "quadratic residue" mod p , i.e. there is an integer a such that $a^2 \equiv 2$ modulo p . [Hint: show there is an $\varepsilon \in \mathbb{Z}_p$ with $\varepsilon^4 = -1$, and $\alpha^2 = 2$ for $\alpha = \varepsilon + \varepsilon^{-1}$.]

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Algebra General Exam August 21, 1999

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This exam is worth 100 points. The number of points each part of a problem is worth is indicated in brackets. If you can't prove part of some problem, you can still use the result of that part in proving subsequent parts.

(1)[10] Let P be a p -Sylow subgroup of a finite group G for some prime p dividing the group order. We know by Sylow theory that the index of the normalizer of P is congruent to 1 modulo p ,

$$[G : N_G(P)] \equiv 1 \pmod{p}$$

Show that the same holds for any larger subgroup M : if $G \supseteq M \supseteq N_G(P)$ then

$$[G : M] \equiv 1 \pmod{p}$$

[Hint: show $N_M(P) = N_G(P)$.]

(2)[20] (a)[5] Show that any finite group G has a faithful action on some set S of cardinality $|S| = |G|$.

From now on (in parts b, c) assume that G is a finite p -group for a prime p , having the property that it has a unique subgroup G_p of order p . [The quaternion group is such a group, with $p = 2$ and $G_2 = \{1, -1\}$.]

(b)[5] Show that G_p is invariant under all endomorphisms of G , $f(G_p) \subseteq G_p$ for all homomorphisms $f : G \rightarrow G$.

(c)[10] Show that G_p "needs room" in order to act: whenever G acts on a finite set S of size $|S| < |G|$, the subgroup G_p acts trivially. Conclude that G can only act faithfully on sets of size $\geq |G|$.

(3)[10] The exponent e of a group G is defined as the smallest positive integer k such that $x^k = 1$ for all $x \in G$; for an abelian group, in additive notation this is the smallest k such that $kx = 0$ for all elements.

If $n_1, n_2, \dots, n_r$ are the invariant factors of a finite abelian group A (so $n_r \mid n_{r-1} \mid \dots \mid n_2 \mid n_1$), prove that A has exponent $e = n_1$, and has an element of order precisely e . Conclude that A has an element of order m iff m divides the largest invariant factor n_1 .

Algebra General Exam January 17, 2000

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This exam is worth 100 points. The number of points each part of a problem is worth is indicated in brackets. If you can't prove part of some problem, you can still use the result of that part in proving subsequent parts.

- (1) [15] (a)[5] Define the trace of an $n \times n$ real matrix, and show that the trace of the product XY of two real $n \times n$ matrices is the same as the trace of YX .
- (b)[5] Explain why the trace of such a matrix is the sum of all the eigenvalues (possibly complex, with the appropriate multiplicities) of the matrix.
- (c)[5] Explain how you would define the trace of an abstract linear operator on any finite-dimensional real vector space V (i.e., a linear transformation from V to V).
- (2)[10] Let K be a field of characteristic zero, $M_n(K)$ the $n \times n$ matrices over K .
- (a)[5] Determine all linear functionals $f : M_n(K) \rightarrow K$ which are symmetric in the sense that $f(ab) = f(ba)$ for all matrices $a, b \in M_n(K)$.
- (a)[5] Determine all linear functionals $f : M_n(K) \rightarrow K$ which are invariant in the sense that $f(gag^{-1}) = f(a)$ for all invertible $g \in GL_n(K)$.
- (3)[15] (a)[5] Let A be an abelian group, and let f and g be any two endomorphisms of A (group homomorphisms of A into itself). Let $B := \text{Fix}(fg) = \{a \in A \mid f(g(a)) = a\}$, $C := \text{Fix}(gf) = \{a \in A \mid g(f(a)) = a\}$. Show that B and C are isomorphic subgroups of A .
- (b)[10] How many abelian groups (up to isomorphism) are there of order 1000 which have no elements whose orders are larger than 35?
- (4)[10] (a)[5] If G is a group containing a cyclic normal subgroup N , show that $gn = ng$ for all n in N and all g in the commutator subgroup of G .
- (b)[5] Suppose that N_1, N_2, N_3 are three normal subgroups of a group G with the properties that for distinct i, j always $N_i \cap N_j = 1$, $N_i N_j = G$. Show that all three subgroups N_i are isomorphic, and that G is abelian. Give an example of such an abelian group of order 4.
- (5)[10] Let G be a finite group such that every element commutes with its conjugates (for any $g, h \in G$ the elements h and ghg^{-1} commute).
- (a)[5] Show that any Sylow subgroup of such a G is normal. [Hard!]
- (b)[5] Explain why the group of quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$ is such a group G .
- (6)[10] Given a finite field K , show there exists a polynomial $f(x, y) \in K[x, y]$ (in which both variables actually appear) for which the equation $f(x, y) = 0$ has no solutions in $K \times K$.
- (7)[15] Suppose that f is an irreducible polynomial of degree n with rational coefficients. Let K be a splitting field for f over the rationals $\mathbb{Q}$, and let $r_1, \dots, r_n$ be the roots of f in K , with

$$a = \prod_{i < j} (r_i - r_j).$$

(a) [5] Show that the roots of f are distinct.

(b)[5] If the product a is not rational, show the Galois group $\text{Gal}(K/\mathbb{Q})$ contains an element which yields an odd permutation of the roots.

(c)[5] If the product a is not rational, show that K contains at least one quadratic subfield.

(8)[15] Let ζ be primitive 3rd root of unity, $K = \mathbb{Q}$, and $L = K\left(\sqrt[3]{2}\right)$.

(a)[10] Show that L/K is a Galois extension, and determine its Galois group $G := \text{Gal}(L/K)$.

(b)[5] Considering L as vector space over K , determine all K -linear functionals $f : L \rightarrow K$ which are G -invariant (i.e. $f(\sigma(a)) = f(a)$ for all $\sigma \in G$ and all $a \in L$).

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- Find all nonconstant polynomials $p(x) \in \mathbb{C}[x]$ satisfying $p(p(p(x))) = p(x)$.
 - Find all rational functions $\varphi(x) \in \mathbb{C}(x)$ satisfying $\varphi(2x) = \varphi(x)$.
- Show that if for a matrix $A \in M_n(K)$ (where K is a field) there exists a nonzero matrix $B \in M_n(K)$ such that $AB = 0$, then there also exists a nonzero matrix $C \in M_n(K)$ such that $CA = 0$ (in other words, in $M_n(K)$ the set of all left zero divisors coincides with the set of all right zero divisors; notice that $AB = 0$ does not always imply $BA = 0$ - find such examples!)
- Let G be a group. Show that if, for any three elements $x, y, z \in G$, at least two of them commute, then G itself is abelian. (Hint. Use centralizers.)
- Show that a finite group is noncommutative if and only if it has an irreducible complex representation of dimension > 1 .
- Let R be a commutative ring, $\mathfrak{a}_1, \dots, \mathfrak{a}_n$ be ideals of R . If for all $i \neq j$ we have $\mathfrak{a}_i + \mathfrak{a}_j = R$ then $\mathfrak{a}_1 \cap \dots \cap \mathfrak{a}_n = \mathfrak{a}_1 \cdot \dots \cdot \mathfrak{a}_n$. (Hint. Induct, starting from $n = 2$.)
- Let $R = \mathbb{C}[x, y] / \mathfrak{a}$ where $\mathbb{C}[x, y]$ is the ring of polynomials in variables x and y , and $\mathfrak{a}$ is the principal ideal of $\mathbb{C}[x, y]$ generated by $p(x, y) = y^2 - x^3$. Show that the ideal of R generated by the images of x and y is not principal.
- Let p be a prime > 2 .
 - Show that in the symmetric group S_p , any two elements of order p are conjugate.
 - Exhibit two nonconjugate elements of order p in the symmetric group S_{p^2} .
 - Are there nonconjugate elements of order p in the group $GL_2(\mathbb{C})$? (Recall that $GL_2(K)$, where K is a field, is the group of all nondegenerate 2×2 -matrices over K with respect to multiplication.)
 - Can you find an element of order p in $GL_2(\mathbb{R})$?
- Is it true that in the ring R of all continuous functions on $[0, 1]$, any element which is not a zero divisor, is invertible?
- Construct a Galois extension of $\mathbb{Q}$ of degree 3.